

Chemistry Chapter 6 — Stoichiometry

Your complete study pack — designed to actually make this fun.

Welcome! Stoichiometry sounds scary (it even *looks* scary — ten letters, three syllables, weird Greek origin). But once you get past the name, it's one of the most beautiful ideas in all of chemistry: **how to count things you can't see**.

What's in this folder?

File	What it's for	When to use it
notes.md	Clean reference notes — all the facts, formulas, and definitions you need	Quick review before a test, or when writing a homework answer
explanations.md	Deep walkthroughs — the "why does this work?" explanations	When something in notes doesn't click. Read slowly.
stories-and-examples.md	Real-world examples and everyday stories that bring each topic to life	To understand WHY this matters outside the exam
history.md	The humans behind the science — Avogadro, Dalton, Lavoisier, and the journey of the mole	For curiosity. Also great cocktail-party facts.
analogies.md	Every topic reframed as something you already understand	When a concept feels abstract — these are your rescue rafts
flashcards.md	60+ Q&A cards organized by difficulty	Daily 10-minute review. Build memory fast.

The 8 topics in this chapter

1. **Empirical vs Molecular Formula** — the two ways to describe a compound
2. **Molecular Mass & Formula Mass** — how heavy is one "unit" of a substance?
3. **The Mole & Avogadro's Number** — chemistry's counting unit (THE big idea)
4. **Gram Atomic/Molecular/Formula Mass (Molar Mass)** — connecting grams to moles
5. **Formulas of Binary Ionic Compounds** — how to write salts like Al_2O_3
6. **Chemical Equations & Balancing** — atoms can't disappear
7. **Ionic Equations** — cutting out the audience, focusing on the action
8. **Molecular vs Structural Formula** — counting atoms vs drawing them

A suggested study plan (1 week)

Day	What to do	Time
Mon	Read explanations.md topics 1-2. Then flashcards for those topics.	40 min
Tue	Read explanations.md topics 3-4 (this is THE MOLE — the core). Re-read the stories for topic 3.	50 min
Wed	Read explanations.md topics 5-6. Try the worked examples yourself before looking.	45 min
Thu	Read explanations.md topics 7-8. Go through analogies.md end-to-end.	40 min
Fri	Full flashcards.md session. Shuffle the order.	30 min
Sat	Read stories-and-examples.md cover to cover. Just enjoy it. No tests.	30 min
Sun	Quick review of notes.md . You're ready.	20 min

One thing to remember before you start

The entire chapter rests on **one single idea**: atoms are too small to count one by one, so chemists invented a counting unit called the **mole**.

Everything else — empirical formulas, molecular masses, balancing equations, ionic equations — is just a different angle on that same idea. If you understand the mole deeply, this whole chapter becomes easy.

So when you're reading, always ask yourself: *how does this connect back to the mole?*

That's your secret weapon.

Good luck. You're going to do great.